Detailed student solutions

Regularized argmax and neural networks as kernel methods

These notes solve all core questions and the extra questions. They are written slowly: the goal is not only to get the answer, but to show why each step is legal.

Contents

1 Question 1: Regularized argmax

We work with

$$\Delta_K = \{\boldsymbol{q} \in \mathbb{R}^K : q_i \geq 0,\ \sum_i q_i = 1\}, \qquad H(\boldsymbol{q}) = -\sum_i q_i \log q_i,$$

with natural logarithms and the convention $0 \log 0 = 0$. The phrase "assume $q_i > 0$" means: restrict attention temporarily to the interior of the simplex so derivatives of $\log q_i$ are well-defined. It does *not* allow negative probabilities. Negative coordinates are outside the feasible set from the beginning.

1a) Hard argmax on the simplex

Let

$$M = \max_{i=1,\dots,K} z_i.$$

For any $\boldsymbol{q} \in \Delta_K$,

$$\langle \boldsymbol{q}, \boldsymbol{z} \rangle = \sum_i q_i z_i \leq \sum_i q_i M = M \sum_i q_i = M.$$

So the value is at most M. This upper bound is achieved by putting all mass on any index i with $z_i = M$, for example $\boldsymbol{q} = e_i$.

Let

$$A = \operatorname*{argmax}_i z_i = \{i : z_i = M\}.$$

Equality holds exactly when $q_i = 0$ for all non-maximizing indices $i \notin A$. Therefore

$$\boxed{\max_{\boldsymbol{q} \in \Delta_K} \langle \boldsymbol{q}, \boldsymbol{z} \rangle = \max_i z_i}$$

and the set of maximizers is

$$\boxed{\operatorname{conv}\{e_i : i \in A\}.}$$

If there is a unique largest score, this set has one element. If several scores tie, the whole face spanned by the tied vertices is optimal.

1b) Deriving softmax from a Lagrangian

We solve

$$\max_{\boldsymbol{q} \in \Delta_K} \{\langle \boldsymbol{q}, \boldsymbol{z} \rangle + H(\boldsymbol{q})\}.$$

First look for an interior stationary point, so $q_i > 0$ for every i. Then the nonnegativity constraints are not active, and we impose only the equality constraint $\sum_i q_i = 1$.

Define the Lagrangian

$$\mathcal{L}(\boldsymbol{q}, \mu) = \sum_i q_i z_i - \sum_i q_i \log q_i - \mu \left(\sum_i q_i - 1 \right).$$

The scalar derivative we need is

$$\frac{d}{dq}[-q \log q] = -(\log q + 1).$$

Thus stationarity gives, for every i,

$$0 = \frac{\partial \mathcal{L}}{\partial q_i} = z_i - \log q_i - 1 - \mu.$$

Rearrange:

$$\log q_i = z_i - 1 - \mu, \qquad q_i = e^{z_i - 1 - \mu} = C e^{z_i},$$

where $C = e^{-1-\mu}$ is the same for every coordinate.

Normalize using $\sum_i q_i = 1$:

$$1 = \sum_i Ce^{z_i} = C\sum_i e^{z_i} \quad \Rightarrow \quad C = \frac{1}{\sum_j e^{z_j}}.$$

Therefore

$$\boxed{q_i^H = \frac{e^{z_i}}{\sum_j e^{z_j}}.}$$

This is the softmax. Notice that this candidate is automatically strictly positive because exponentials are positive. The remaining question is whether this stationary point is the global maximum over the full simplex, including boundary points with some $q_i = 0$. Parts 1c–1e answer that.

1c) Why it was safe to omit the inequalities in the Lagrangian

The short answer is: the inequalities are still part of the problem. We did not allow negative probabilities. We only left the constraints $q_i \geq 0$ out of the Lagrangian while searching for an interior candidate.

1. The candidate satisfies nonnegativity automatically. From 1b,

$$q_i = Ce^{z_i}, \qquad C = \frac{1}{\sum_j e^{z_j}} > 0.$$

Since exponentials are strictly positive, every coordinate satisfies

$$q_i > 0.$$

So the candidate is not merely feasible; it is strictly inside the simplex. In particular, it cannot be negative. Negative coordinates were never feasible anyway, because the domain of the problem is Δ_K.

2. If we included inequality multipliers, they would vanish. Write the inequalities as

$$-q_i \leq 0.$$

One convenient maximization Lagrangian is

$$\mathcal{L}(q, \mu, \nu) = \sum_i q_i z_i - \sum_i q_i \log q_i - \mu\left(\sum_i q_i - 1\right) + \sum_i \nu_i q_i, \qquad \nu_i \geq 0.$$

Stationarity would become

$$z_i - \log q_i - 1 - \mu + \nu_i = 0.$$

Complementary slackness says

$$\nu_i q_i = 0.$$

Since the optimizer is interior, $q_i > 0$, complementary slackness forces

$$\nu_i = 0 \quad \text{for all } i.$$

Thus the full KKT conditions reduce exactly to the equality-only Lagrangian from 1b.

3. Why can no entropy optimum have $q_i = 0$? The derivative of $-q \log q$ is

$$-(\log q + 1).$$

As $q \downarrow 0$, this derivative tends to $+\infty$. Intuitively, putting a tiny amount of mass on an empty coordinate gives an enormous entropy gain. More concretely, if $q_i = 0$, moving ε mass from a positive coordinate to coordinate i creates the term

$$-\varepsilon \log \varepsilon,$$

whose gain per unit mass satisfies

$$\frac{-\varepsilon \log \varepsilon}{\varepsilon} = -\log \varepsilon \to \infty.$$

A finite score difference $z_i - z_j$ cannot compensate for this. Therefore no boundary point can maximize the entropy-regularized objective.

4. Why this matters later. For the quadratic regularizer, the derivative at the boundary is finite. There is no entropy barrier. The equality-only stationary point can leave the interval $[0, 1]$ in the binary case, so the inequality constraints become active and we must handle them. This is exactly why sparsemax can produce exact zeros, while softmax cannot.

Teaching takeaway.

A common recipe is: drop constraints you expect to be inactive, solve the simpler system, then check feasibility and global optimality. Entropy is a special case where the check always succeeds: the solution is strictly positive and the global proofs below compare against the whole simplex.

1d) First global proof: concavity and boundary behavior

We first show the objective is strictly concave in the interior. Consider the scalar function

$$h(q) = -q \log q, \qquad q > 0.$$

Then

$$h'(q) = -(\log q + 1), \qquad h''(q) = -\frac{1}{q} < 0.$$

So h is strictly concave on $(0, \infty)$. Since

$$H(\boldsymbol{q}) = \sum_i h(q_i),$$

entropy is strictly concave on the relative interior of the simplex. The linear score $\langle \boldsymbol{q}, \boldsymbol{z} \rangle$ is both concave and convex, so adding it preserves strict concavity. Thus there can be at most one interior maximizer.

Now we check that the maximum cannot occur on the boundary. Suppose $q_i = 0$ for some coordinate i. Since $\sum_j q_j = 1$, there is some j with $q_j > 0$. Move a small amount $\varepsilon > 0$ of mass from coordinate j to coordinate i:

$$q_i(\varepsilon) = \varepsilon, \qquad q_j(\varepsilon) = q_j - \varepsilon.$$

The linear part changes only by order ε:

$$\varepsilon(z_i - z_j).$$

The entropy contribution from the new coordinate is

$$-\varepsilon \log \varepsilon.$$

As $\varepsilon \downarrow 0$, this term is positive and has infinite directional derivative because

$$\frac{-\varepsilon \log \varepsilon}{\varepsilon} = -\log \varepsilon \to \infty.$$

So moving slightly into the simplex increases the objective. Hence no boundary point can be a maximizer.

Therefore the maximizer is interior. Since the objective is strictly concave in the interior, the stationary point from 1b is the unique global maximizer.

Conclusion.

$$\boxed{\operatorname*{argmax}_{\boldsymbol{q} \in \Delta_K} \{\langle \boldsymbol{q}, \boldsymbol{z} \rangle + H(\boldsymbol{q})\} = \operatorname{softmax}(\boldsymbol{z}).}$$

1e) Second global proof: KL / Gibbs identity

Let

$$p_i = \frac{e^{z_i}}{Z}, \qquad Z = \sum_j e^{z_j}.$$

Then

$$\log p_i = z_i - \log Z, \qquad \text{hence} \qquad z_i = \log p_i + \log Z.$$

For any $\boldsymbol{q} \in \Delta_K$,

$$\begin{aligned}
\langle \boldsymbol{q}, \boldsymbol{z} \rangle + H(\boldsymbol{q}) &= \sum_i q_i z_i - \sum_i q_i \log q_i \\
&= \sum_i q_i (\log p_i + \log Z) - \sum_i q_i \log q_i \\
&= \log Z \sum_i q_i + \sum_i q_i \log p_i - \sum_i q_i \log q_i \\
&= \log Z - \sum_i q_i \log \frac{q_i}{p_i} \\
&= \boxed{\log Z - D_{\mathrm{KL}}(\boldsymbol{q} \| \boldsymbol{p}).}
\end{aligned}$$

Since $D_{\mathrm{KL}}(\boldsymbol{q}\|\boldsymbol{p}) \geq 0$, we get

$$\langle \boldsymbol{q}, \boldsymbol{z} \rangle + H(\boldsymbol{q}) \leq \log Z.$$

Equality holds if and only if $\boldsymbol{q} = \boldsymbol{p}$. Therefore the unique maximizer is

$$\boxed{\boldsymbol{p} = \mathrm{softmax}(\boldsymbol{z})}$$

and the optimal value is

$$\boxed{\log \sum_i e^{z_i}.}$$

This is the Gibbs variational principle in finite-dimensional form.

1f) Quadratic regularization in two classes

Let $K = 2$, $z = (t, 0)^\top$, and $q = (p, 1-p)^\top$. The objective becomes

$$\begin{aligned}
\psi_t(p) &= \langle q, z \rangle - \frac{1}{2}\|q\|_2^2 \\
&= pt - \frac{1}{2}\left(p^2 + (1-p)^2\right) \\
&= pt - \frac{1}{2}(2p^2 - 2p + 1) \\
&= -p^2 + (t+1)p - \frac{1}{2},
\end{aligned}$$

with constraint $0 \leq p \leq 1$.

The unconstrained derivative is

$$\psi_t'(p) = -2p + t + 1.$$

Setting this to zero gives

$$p_{\mathrm{unc}} = \frac{1+t}{2}.$$

Now enforce $0 \leq p \leq 1$ by clipping:

$$\boxed{p^Q(t) = \begin{cases} 0, & t \leq -1, \\ \dfrac{1+t}{2}, & -1 < t < 1, \\ 1, & t \geq 1. \end{cases}}$$

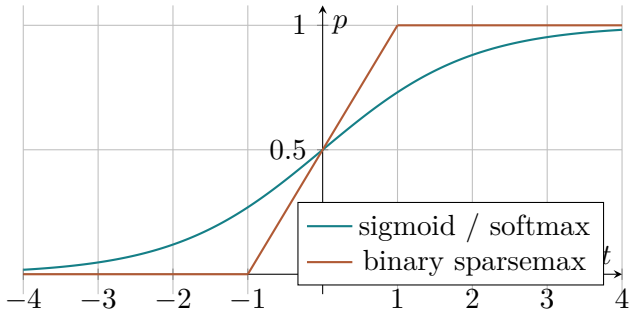
The entropy-regularized two-class solution is

$$p^H(t) = \frac{e^t}{e^t + 1} = \frac{1}{1 + e^{-t}}.$$

So $p^H(t)$ is the sigmoid. It is strictly between 0 and 1 for every finite t, while $p^Q(t)$ reaches exact 0 for $t \le -1$ and exact 1 for $t \ge 1$.

Active constraints.

For $t < -1$, the lower constraint $p \ge 0$ is active. For $-1 < t < 1$, no bound is active. For $t > 1$, the upper constraint $p \le 1$ is active. At $t = \pm 1$, a constraint is active but its multiplier can be zero.



2 Question 1 extras

1g) Temperature and limits

Solve

$$q^{(\tau)}(z) = \arg\max_{q \in \Delta_K} \{\langle q, z \rangle + \tau H(q)\}.$$

The Lagrangian is

$$\mathcal{L}(q, \mu) = \sum_i q_i z_i - \tau \sum_i q_i \log q_i - \mu \left(\sum_i q_i - 1 \right).$$

Stationarity gives

$$0 = z_i - \tau(\log q_i + 1) - \mu.$$

Hence

$$\log q_i = \frac{z_i}{\tau} - 1 - \frac{\mu}{\tau}, \qquad q_i = C e^{z_i/\tau}.$$

After normalization,

$$\boxed{q_i^{(\tau)}(z) = \frac{e^{z_i/\tau}}{\sum_j e^{z_j/\tau}}.}$$

As $\tau \downarrow 0$, the distribution concentrates on the maximum-score indices. If the maximum is unique, it converges to the corresponding vertex. If the maximum is tied over a set A, then the limit is uniform on A:

$$q_i^{(\tau)} \to \begin{cases} 1/|A|, & i \in A, \\ 0, & i \notin A. \end{cases}$$

As $\tau \to \infty$, all ratios $z_i/\tau \to 0$, so $e^{z_i/\tau} \to 1$, and

$$q_i^{(\tau)} \to \frac{1}{K}.$$

1h) General sparsemax via KKT

Consider

$$q^Q(z) = \arg\max_{q\in\Delta_K} \left\{ \langle q, z\rangle - \frac{1}{2}\|q\|_2^2 \right\}.$$

Complete the square:

$$\langle q, z\rangle - \frac{1}{2}\|q\|^2 = -\frac{1}{2}\|q - z\|^2 + \frac{1}{2}\|z\|^2.$$

The second term is constant in q, so the maximization is equivalent to

$$\boxed{q^Q(z) = \arg\min_{q\in\Delta_K} \frac{1}{2}\|q - z\|^2.}$$

Thus sparsemax is Euclidean projection onto the simplex.

Use constraints $q_i \geq 0$, written as $-q_i \leq 0$, and $\sum_i q_i = 1$. For the minimization form, the Lagrangian is

$$\mathcal{L}(q, \theta, \mu) = \frac{1}{2}\sum_i (q_i - z_i)^2 + \theta\left(\sum_i q_i - 1\right) - \sum_i \mu_i q_i, \qquad \mu_i \geq 0.$$

Stationarity:

$$0 = q_i - z_i + \theta - \mu_i.$$

Complementary slackness:

$$\mu_i q_i = 0.$$

If $q_i > 0$, then $\mu_i = 0$, so $q_i = z_i - \theta$. If $q_i = 0$, stationarity gives $\mu_i = \theta - z_i \geq 0$, so $z_i - \theta \leq 0$. Combining both cases:

$$\boxed{q_i = [z_i - \theta]_+.}$$

The threshold θ is chosen so that $\sum_i q_i = 1$:

$$\boxed{\sum_i [z_i - \theta]_+ = 1.}$$

The left-hand side is continuous and decreasing in θ, so the threshold is unique.

For $K = 2$, $z = (t, 0)$:

$$q_1 = [t - \theta]_+, \qquad q_2 = [-\theta]_+.$$

If both are positive, $t - \theta - \theta = 1$, so $\theta = (t-1)/2$, valid only for $-1 < t < 1$. If only q_2 is positive, $\theta = -1$, valid for $t \leq -1$. If only q_1 is positive, $\theta = t - 1$, valid for $t \geq 1$. This recovers 1f.

1i*) Softmax Jacobian and sensitivity

Let

$$F(z) = \log \sum_i e^{z_i}, \qquad q_i = \frac{e^{z_i}}{\sum_j e^{z_j}}.$$

Then

$$\frac{\partial F}{\partial z_i} = \frac{e^{z_i}}{\sum_j e^{z_j}} = q_i,$$

so $\nabla F = q$.

Differentiate q_i with respect to z_j:

$$\frac{\partial q_i}{\partial z_j} = q_i(\delta_{ij} - q_j).$$

Therefore

$$\boxed{J(z) = \nabla^2 F(z) = \mathrm{diag}(q) - qq^\top.}$$

For any vector v,

$$\begin{aligned} v^T J v &= \sum_i q_i v_i^2 - \left(\sum_i q_i v_i\right)^2 \\ &= \mathrm{Var}_{I\sim q}(v_I). \end{aligned}$$

A variance is bounded by one quarter of the squared range:

$$\mathrm{Var}(v_I) \le \frac{(\max_i v_i - \min_i v_i)^2}{4}.$$

For $\|v\|_2 = 1$, the largest possible coordinate difference is at most $\sqrt{2}$. Hence

$$v^T J v \le \frac{2}{4} = \frac{1}{2}.$$

So $\|J(z)\|_{2\to 2} \le 1/2$.

Now integrate the derivative along the line segment from z' to z:

$$q(z) - q(z') = \int_0^1 J(z' + s(z - z'))(z - z')\, ds.$$

Taking norms,

$$\|q(z) - q(z')\|_2 \le \int_0^1 \frac{1}{2}\|z - z'\|_2\, ds = \frac{1}{2}\|z - z'\|_2.$$

The constant is best possible: for $K = 2$, at $z = (0, 0)$,

$$J = \begin{pmatrix} 1/4 & -1/4 \\ -1/4 & 1/4 \end{pmatrix}$$

has largest eigenvalue $1/2$. Finite differences near this point approach the ratio $1/2$.

3 Question 2: Training a neural network's output layer

The model is

$$f_a(x) = \sum_{r=1}^{m} a_r \sigma(w_r^T x),$$

where the hidden weights w_r are fixed and only the output weights a_r are trained. This is the simplest setting in which a neural network becomes a kernel method.

2a) Feature map

Define

$$\phi_r(x) = \sigma(w_r^T x), \qquad r = 1, \dots, m,$$

and

$$\phi(x) = (\phi_1(x), \dots, \phi_m(x))^\top \in \mathbb{R}^m.$$

Then

$$a^T \phi(x) = \sum_{r=1}^{m} a_r \sigma(w_r^T x) = f_a(x).$$

So the network is a linear model in the nonlinear feature vector $\phi(x)$.

2b) Stationarity of ridge regression

Let

$$\phi_i = \phi(x_i), \qquad p_i = f_a(x_i) = a^T \phi_i.$$

The objective is

$$J(a) = \frac{1}{2} \sum_i (p_i - y_i)^2 + \frac{\lambda}{2} \|a\|^2.$$

Differentiate:

$$\nabla_a \frac{1}{2}(p_i - y_i)^2 = (p_i - y_i) \nabla_a p_i = (p_i - y_i)\phi_i.$$

Also

$$\nabla_a \frac{\lambda}{2} \|a\|^2 = \lambda a.$$

Therefore

$$\nabla_a J(a) = \sum_i (f_a(x_i) - y_i)\phi_i + \lambda a.$$

At the optimum $a^\star$,

$$\boxed{\sum_i (f_{a^\star}(x_i) - y_i)\phi_i + \lambda a^\star = 0.}$$

The solution is unique because the Hessian is $\sum_i \phi_i \phi_i^T + \lambda I$, which is positive definite when $\lambda > 0$.

2c) The solution lies in the data-feature span

Rearrange the stationarity condition:

$$\lambda a^\star = \sum_i (y_i - f_{a^\star}(x_i))\phi_i.$$

So

$$\boxed{a^\star = \sum_i \alpha_i \phi_i, \qquad \alpha_i = \frac{y_i - f_{a^\star}(x_i)}{\lambda}.}$$

The coefficients still involve the unknown fitted predictions. That is okay. This part proves a structural fact: the optimal weight vector lies in the span of the training feature vectors. Part 2e turns this into a closed linear system for α.

2d) The kernel appears

For any input x,

$$\begin{aligned} f_{a^\star}(x) &= (a^\star)^T \phi(x) \\ &= \left(\sum_i \alpha_i \phi(x_i) \right)^T \phi(x) \\ &= \sum_i \alpha_i \phi(x_i)^T \phi(x). \end{aligned}$$

Define

$$\boxed{k(x, x') = \phi(x)^T \phi(x') = \sum_{r=1}^{m} \sigma(w_r^T x)\sigma(w_r^T x').}$$

Then

$$\boxed{f_{a^\star}(x) = \sum_i \alpha_i k(x_i, x).}$$

To prove positive semidefiniteness, take any points $x_1, \ldots, x_N$ and coefficients $c_1, \ldots, c_N$:

$$\begin{aligned} \sum_{i,j} c_i c_j k(x_i, x_j) &= \sum_{i,j} c_i c_j \phi(x_i)^T \phi(x_j) \\ &= \left\| \sum_i c_i \phi(x_i) \right\|^2 \\ &\geq 0. \end{aligned}$$

So k is a valid kernel.

2e) Kernel system and practical payoff

Let

$$K_{ij} = k(x_i, x_j), \qquad \alpha = (\alpha_i)_i, \qquad y = (y_i)_i.$$

On the training set,

$$p_i = f_{a^\star}(x_i) = \sum_j \alpha_j k(x_j, x_i).$$

Since K is symmetric, this is

$$p = K\alpha.$$

From 2c,

$$\alpha_i = \frac{y_i - p_i}{\lambda},$$

so in vector form,

$$\alpha = \frac{1}{\lambda}(y - K\alpha).$$

Multiply by λ:

$$\lambda\alpha = y - K\alpha.$$

Therefore

$$\boxed{(K + \lambda I)\alpha = y.}$$

Correcting the displayed formula to remove the stray character:

$$\boxed{(K + \lambda I)\alpha = y.}$$

Thus

$$\boxed{\alpha = (K + \lambda I)^{-1} y}$$

and

$$\boxed{f_{a^\star}(x) = k_x^T (K + \lambda I)^{-1} y,}$$

where $(k_x)_i = k(x_i, x)$.

The direct primal solution solves an $m \times m$ system:

$$(\Phi^T \Phi + \lambda I_m) a = \Phi^T y.$$

The kernel solution solves an $N \times N$ system:

$$(K + \lambda I_N) \alpha = y.$$

If $m \ll N$, the primal system is usually cheaper. If $N \ll m$, the kernel system is usually cheaper. The kernel view is especially powerful when the feature dimension is huge or infinite but the kernel can be computed directly.

If hidden weights were trained too, then $\phi(x)$ would change during training. The fixed kernel $k(x, x') = \phi(x)^T \phi(x')$ would no longer describe the whole optimization. That is why full networks need another argument, such as linearization around an initialization.

4 Question 2 extras

2f) Concrete ReLU kernel

Here

$$w_1 = 1, \qquad w_2 = -1, \qquad \sigma(u) = \max(0, u).$$

So

$$\phi(x) = \begin{pmatrix} \max(0, x) \\ \max(0, -x) \end{pmatrix}.$$

The kernel is

$$k(x, x') = \max(0, x) \max(0, x') + \max(0, -x) \max(0, -x').$$

If x and x' are both positive, this equals xx'. If both are negative, it equals $(-x)(-x') = xx'$, which is again positive. If they have opposite signs, one coordinate of each feature vector is zero in opposite places, so the inner product is zero. Thus

$$\boxed{k(x, x') = \max(0, xx').}$$

Therefore

$$k(1, 2) = 2, \qquad k(1, -2) = 0.$$

Inputs with opposite signs activate different ReLU features and are orthogonal under this kernel.

2g*) Gradient descent in kernel space

Let $\Phi \in \mathbb{R}^{N \times m}$ have rows $\phi(x_i)^T$. Then

$$p^{(s)} = \Phi a^{(s)}, \qquad K = \Phi \Phi^T.$$

Gradient descent gives

$$a^{(s+1)} = a^{(s)} - \eta \left(\Phi^T (p^{(s)} - y) + \lambda a^{(s)} \right).$$

Multiply by Φ:

$$\begin{aligned} p^{(s+1)} &= \Phi a^{(s+1)} \\ &= \Phi a^{(s)} - \eta \left(\Phi \Phi^T (p^{(s)} - y) + \lambda \Phi a^{(s)} \right) \\ &= p^{(s)} - \eta \left(K (p^{(s)} - y) + \lambda p^{(s)} \right). \end{aligned}$$

So the prediction dynamics are

$$\boxed{p^{(s+1)} = p^{(s)} - \eta\left(K(p^{(s)} - y) + \lambda p^{(s)}\right).}$$

Starting from $a^{(0)} = 0$, the first update is a linear combination of training features. If $a^{(s)}$ is in their span, then the gradient term $\Phi^T(p^{(s)} - y)$ is also in their span, and $\lambda a^{(s)}$ is in their span. Hence every iterate remains in the training-feature span.

If the dynamics converge to p^∞, the update term must vanish:

$$K(p^\infty - y) + \lambda p^\infty = 0.$$

Thus

$$(K + \lambda I)p^\infty = Ky.$$

Using $\alpha = (K + \lambda I)^{-1}y$,

$$\boxed{p^\infty = K(K + \lambda I)^{-1}y = K\alpha.}$$

So gradient descent converges to the same fitted training predictions as kernel ridge regression, assuming a stable step size and convergence.

2h*) Random features limit

With scaling,

$$f_a(x) = \frac{1}{\sqrt{m}}\sum_{r=1}^{m} a_r \sigma(w_r^T x).$$

The features are

$$\phi_r(x) = \frac{1}{\sqrt{m}}\sigma(w_r^T x).$$

Therefore the finite-width kernel is

$$\boxed{k_m(x, x') = \frac{1}{m}\sum_{r=1}^{m} \sigma(w_r^T x)\sigma(w_r^T x').}$$

If $w_r \overset{\text{iid}}{\sim} \rho$, then by the law of large numbers,

$$k_m(x, x') \to \boxed{\mathbb{E}_{w\sim\rho}[\sigma(w^T x)\sigma(w^T x')]}$$

for fixed x, x' as $m \to \infty$. This gives a deterministic limiting kernel.

The usefulness is that we can reason about an infinite-dimensional feature map through a kernel function. We do not need to explicitly store infinitely many random features.

2i**) Full-network linearization / tangent kernel

Let θ contain all trainable weights and fix θ_0. Define

$$g(x) = \nabla_\theta f_{\theta_0}(x).$$

The first-order model is

$$\widetilde{f}_\delta(x) = f_{\theta_0}(x) + g(x)^T\delta.$$

Fit δ with squared loss plus $\lambda\|\delta\|^2/2$. This is ridge regression in features $g(x)$, but the targets are shifted by the initial prediction. Define residual targets

$$r_i = y_i - f_{\theta_0}(x_i).$$

Then the fitted correction is kernel ridge regression with kernel

$$\boxed{k_{\theta_0}(x, x') = g(x)^T g(x') = \nabla_\theta f_{\theta_0}(x)^T \nabla_\theta f_{\theta_0}(x').}$$

The final prediction is

$$\boxed{\widetilde{f}_{\delta^\star}(x) = f_{\theta_0}(x) + k_x^T (K + \lambda I)^{-1} r.}$$

For the two-layer network

$$f_\theta(x) = \frac{1}{\sqrt{m}} \sum_{r=1}^{m} a_r \sigma(w_r^T x),$$

the derivative with respect to a_r is

$$\frac{\partial f}{\partial a_r}(x) = \frac{1}{\sqrt{m}} \sigma(w_r^T x).$$

The derivative with respect to w_r is

$$\nabla_{w_r} f(x) = \frac{1}{\sqrt{m}} a_r \sigma'(w_r^T x) x.$$

Thus the tangent kernel separates into output-weight and hidden-weight contributions:

$$\boxed{k_{\theta_0}(x, x') = \frac{1}{m} \sum_{r=1}^{m} \sigma(w_r^T x)\sigma(w_r^T x') + \frac{1}{m}(x^T x') \sum_{r=1}^{m} a_r^2 \sigma'(w_r^T x)\sigma'(w_r^T x')}$$

with all a_r, w_r evaluated at θ_0.

If all output weights are initialized at zero, $a_r = 0$, then the hidden-weight contribution vanishes. Infinitesimal hidden-weight changes do not affect the output to first order when the output layer is exactly zero.